II-Probabilistic Approaches Probability, Marginal and Conditional Distributions

Foundations of Neuro-Symbolic AI

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Outline

- Handling uncertainties
- Probability distributions
- Marginal distributions
- Conditional distributions
- Connections with logics

Toy accounting example

Let us imagine that each possible assignment to is of equal frequency.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \; [X_{A1}, X_{A2}, X_F]$$

We can ask the following questions: Is the account Account 1 or the account Account 2 more likely to be booked?

Since logic can only handle certainty, it cannot answer this question.

However, we can intuitively assign probabilities to the states:

$$\begin{pmatrix} 0 & \frac{1}{3} \\ \frac{1}{3} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ \frac{1}{3} & 0 \end{pmatrix} \quad [X_{A1}, X_{A2}, X_F]$$

There are two worlds where Account 1 is booked, and one world where Account 2 is booked, thus

$$\mathbb{P}[X_{A1} = 1] = \frac{2}{3} \quad > \quad \mathbb{P}[X_{A2} = 1] = \frac{1}{3}.$$

Uncertainty: Summarizing the unknown

Uncertainties need to be handled:

- Not all variables are measured.
  Example: When we do not know the value of Feature, then we do not know whether Account 1 or Account 2 have to be booked.
- Parts of the models are not known.
  Example: Even if Feature is known to be False, then we do not know whether Account 1 or Account 2 have to be booked.

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Probability distributions

Instead of storing 0/1 for possibilities, we can store probabilities (in the interval $[0, 1]$) in the coordinates of a tensor.
The total probability of all states must be 1 and can be calculated by contracting all legs of the tensor since

$$\sum_{x_{[d]}} \mathbb{P}\left[X_{[d]} = x_{[d]}\right] = \left\langle \mathbb{P}\left[X_{[d]}\right] \right\rangle_{[\varnothing]} .$$

Definition (Probability Tensor)

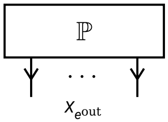
Let there be a factored system $X_{[d]}$ defined by variables X_k taking values in $[m_k]$. A probability distribution over the states of $X_{[d]}$ is a tensor $\mathbb{P}\left[X_{[d]}\right]$ with coordinates in the interval $[0, 1]$ such that

$$\left\langle \mathbb{P}\left[X_{[d]}\right] \right\rangle_{[\varnothing]} = 1 .$$

Directed hyperedges

Let us introduce directed hyperedges: Instead of just being subsets of the variables, directed hyperedges are pairs of incoming variables $e^{\mathrm{in}} \subset \mathcal{V}$ and outgoing variables $e^{\mathrm{out}} \subset \mathcal{V}$.

To highlight the normalization constraint we understand probability distributions as tensors on directed hyperedges, where all variables are outgoing:



Incoming variables will be used later to highlight conditioning.

Outline

Example: Being at a dentist

We are reasoning about a factored system with three variables:

- Toothache $x_0 \in [2]$, whether your tooth hurts
- Cavity $x_1 \in [2]$, whether there is a cavity in your tooth
- Catch $x_2 \in [2]$, whether the dentist catches in your tooth

Example: Being at a dentist

Let us take the following probability distribution over the states of the system:

Toothache x_0	Cavity x_1	Catch x_2	$\mathbb{P}[X_0 = x_0, X_1 = x_1, X_2 = x_2]$
0	0	0	0.576
0	0	1	0.144
0	1	0	0.008
0	1	1	0.072
1	0	0	0.064
1	0	1	0.016
1	1	0	0.012
1	1	1	0.108

We align the probability values in the coordinates of an order-3 tensor:

$$\begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix}^{\begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix}} [X_0, X_1, X_2]$$

Marginal distributions

Example: What is the relation between Toothache and Cavity?

- The variable Catch is not relevant for this question, thus we want to marginalize over it.
- To this end, we sum the probability of states differing in the value of Catch only.

This amounts to a contraction of the probability distribution tensor closing the variable X_2:

$$\begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} \quad [X_0, X_1, X_2]$$

$$\downarrow$$

$$\begin{pmatrix} * & * \\ * & * \end{pmatrix} \qquad [X_0, X_1]$$

Marginal distributions

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$$\Big\downarrow$$

$$\begin{pmatrix} 0.72 & * \\ * & * \end{pmatrix} \qquad [X_0, X_1]$$

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$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} \; [X_0, X_1]$$

Marginal distributions

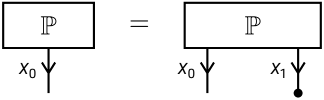
Summation of probabilities of states are done by contractions of the probability distribution tensor.

Definition (Marginal distribution)

Given a probability distribution $\mathbb{P}\left[X_{[d]}\right]$ and a subset $\mathcal{U} \subset [d]$, the marginal distribution of the variables $X_{\mathcal{U}}$ is the distribution

$$\mathbb{P}\left[X_{\mathcal{U}}\right] = \left\langle \mathbb{P}\left[X_{[d]}\right] \right\rangle_{\left[X_{\mathcal{U}}\right]} .$$

Example: The marginal distribution of $\mathcal{U} = \{X_0\}$ given a joint distribution of two variables X_0 and X_1 is the contraction



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Example: Conditional distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states:

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \quad \longrightarrow \quad \begin{pmatrix} * & * \\ * & * \end{pmatrix} \; [X_0 | X_1]$$

This is called a conditional distribution.

Example: Conditional distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states:

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \longrightarrow \begin{pmatrix} \frac{0.72}{0.72+0.08} & \frac{0.08}{0.72+0.08} \\ * & * \end{pmatrix} [X_0 | X_1]$$

This is called a conditional distribution.

Example: Conditional distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states:

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \quad \longrightarrow \quad \begin{pmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{pmatrix} [X_1 | X_0]$$

This is called a conditional distribution.

Normalizations

More general, conditionalization amounts to normalization of probability tensors.

Definition (Normalization)

The normalization of a tensor $\tau\left[X_{[d]}\right]$ with respect to disjoint subsets $\mathcal{V}^{\mathrm{out}}, \mathcal{V}^{\mathrm{in}} \subset [d]$ is the tensor

$$\left\langle \tau\left[X_{[d]}\right] \right\rangle_{\left[X_{\mathcal{V}^{\mathrm{out}}} | X_{\mathcal{V}^{\mathrm{in}}}\right]}$$

defined by the coordinates

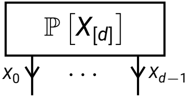
$$\left\langle \tau\left[X_{[d]}\right] \right\rangle_{\left[X_{\mathcal{V}^{\mathrm{out}}}=x_{\mathcal{V}^{\mathrm{out}}} | X_{\mathcal{V}^{\mathrm{in}}}=x_{\mathcal{V}^{\mathrm{in}}}\right]} = \begin{cases} \dfrac{\left\langle \tau\left[X_{[d]}\right] \right\rangle_{\left[X_{\mathcal{V}^{\mathrm{out}}}=x_{\mathcal{V}^{\mathrm{out}}}, X_{\mathcal{V}^{\mathrm{in}}}=x_{\mathcal{V}^{\mathrm{in}}}\right]}}{\left\langle \tau\left[X_{[d]}\right] \right\rangle_{\left[X_{\mathcal{V}^{\mathrm{in}}}=x_{\mathcal{V}^{\mathrm{in}}}\right]}} & \text{if} \quad \left\langle \tau\left[X_{[d]}\right] \right\rangle_{\left[X_{\mathcal{V}^{\mathrm{in}}}=x_{\mathcal{V}^{\mathrm{in}}}\right]} \neq 0 \\ \dfrac{1}{\prod_{v \in \mathcal{V}^{\mathrm{out}}} m_v} & \text{else} \end{cases}$$

Normalized tensors and directed notation

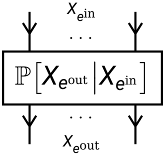
Directed tensors represent normalizations: Each slice of the incoming variables is a normalized tensor, i.e. coordinates summing to 1.
For example a probability distribution is normalized without incoming variables since

$$\left\langle \mathbb{P}\left[X_{[d]}\right] \right\rangle_{\left[X_{[d]}|\varnothing\right]} = \mathbb{P}\left[X_{[d]}\right]$$

This is denoted by



A conditional distribution is invariant under further normalizations (with the same incoming variables) and is thus denoted by

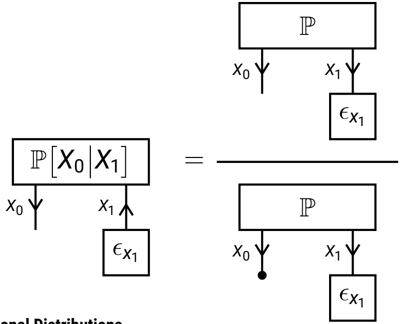


Formal Definition of Conditional Distributions

Definition (Conditional Probability)

Given a distribution $\mathbb{P}\left[X_{[d]}\right]$ and disjoint subsets $\mathcal{V}^{\text{out}}, \mathcal{V}^{\text{in}} \subset [d]$, the distribution of $X_{\mathcal{V}^{\text{out}}}$ conditioned on $X_{\mathcal{V}^{\text{in}}}$ is the normalization

$$\mathbb{P}\left[X_{\mathcal{V}^{\text{out}}} \middle| X_{\mathcal{V}^{\text{in}}}\right] = \left\langle \mathbb{P}\left[X_{[d]}\right] \right\rangle_{\left[X_{\mathcal{V}^{\text{out}}} | X_{\mathcal{V}^{\text{in}}}\right]}$$



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Events and their probability

Events are subsets

$$\mathcal{U} \subset \bigtimes_{k\in[d]} [m_k]$$

of the possible samples. When the leg dimensions are 2 we build the formula

$$f\left[X_{[d]}\right] = \sum_{x_{[d]}\in\mathcal{U}} \epsilon_{x_{[d]}}\left[X_{[d]}\right] .$$

Probability of the event is then the contraction

$$\sum_{x_{[d]}\,:\,f\left[X_{[d]}=x_{[d]}\right]=1} \mathbb{P}\left[X_{[d]}=x_{[d]}\right] = \left\langle \mathbb{P}\left[X_{[d]}\right], f\left[X_{[d]}\right]\right\rangle_{[\varnothing]} .$$

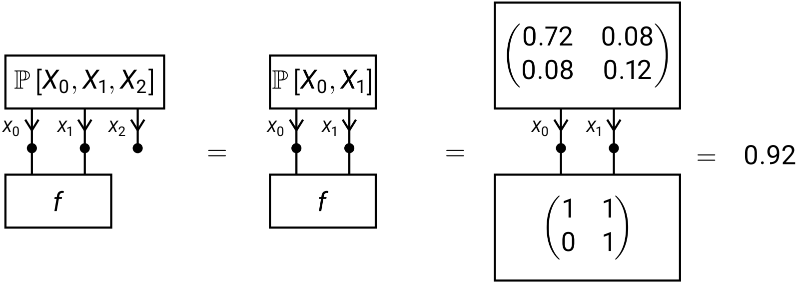
Toothache example

Example: What is the probability that there is a cavity whenever having toothache?

This is the event corresponding with the formula

$$f\,[X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} [X_0, X_1]$$

Since only two variables are affected by this, the probability of the event can be answered by the marginal distribution:



Comparing logics and probability theory

Modelling assumptions

- **Ontological**: What properties does a system have?
  $\rightarrow$ We assume fixed sets of categorical properties.
- **Epistemological**: What can we tell about these properties?
  $\rightarrow$ We associate possibilities or probabilities with states.

Two main frameworks differing in the empistemological assumptions:

- Logics (Possibilities): The traditional mathematical model of intelligence.
- Probability Theory (Probabilities): Handling uncertainties about the systems state.

In both cases, we encode our knowledge in an arrangement of numbers: Tensors with efficient representations by Tensor Networks.